

DSA 8020 R Lab 5: Analysis of covariance and Non-linear Regression

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Analysis of covariance: Salaries for Professors

The 2008-09 nine-month academic salary for Assistant Professors, Associate Professors, and Professors in a college in the U.S. was collected as part of the ongoing effort of the college's administration to monitor salary differences between male and female faculty members.

Load the dataset

Code:

```
library(carData)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
data(Salaries)
head(Salaries)
```

```
##      rank discipline yrs.since.phd yrs.service  sex salary
## 1    Prof          B           19          18 Male 139750
## 2    Prof          B           20          16 Male 173200
## 3 AsstProf          B            4            3 Male  79750
## 4    Prof          B           45          39 Male 115000
## 5    Prof          B           40          41 Male 141500
## 6 AssocProf          B            6            6 Male  97000
```

Description of the variables

- **rank**: a factor with levels Assistant Professor (“AsstProf”); Associate Professor (“AssocProf”); Full Professor (“Prof”)
- **discipline**: a factor with levels A (“theoretical” departments) or B (“applied” departments)
- **yrs.since.phd**: years since her/his PhD
- **sex**: a factor with levels “Female” and “Male”
- **salary**: nine-month salary, in dollars

Exploratory Data Analysis

1. Identify the numerical variables and categorical variables in this data set

Answer:

Numerical Vars

yrs.since.phd, salary

Categorical Vars

rank, discipline, sex

2. Summarize each variable numerically and graphically, briefly describe your findings

Code:

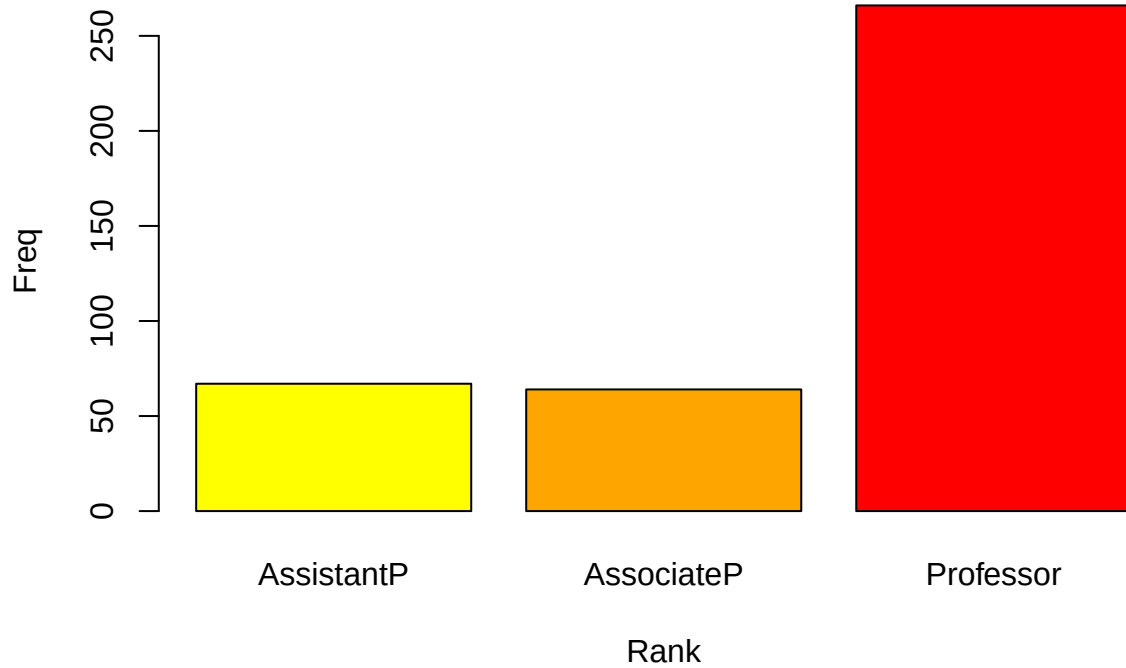
```
vars <- c("rank", "discipline", "yrs.since.phd", "sex", "salary")
data <- Salaries[, vars]
data <- data %>%
  mutate(rank = as.numeric(rank),
         discipline = as.numeric(discipline),
         sex = as.numeric(sex))
```

```
# numerical summary
summary(data)
```

```
##      rank      discipline  yrs.since.phd      sex
##  Min.   :1.000  Min.   :1.000  Min.   : 1.00  Min.   :1.000
##  1st Qu.:2.000  1st Qu.:1.000  1st Qu.:12.00  1st Qu.:2.000
##  Median :3.000  Median :2.000  Median :21.00  Median :2.000
##  Mean   :2.501  Mean   :1.544  Mean   :22.31  Mean   :1.902
##  3rd Qu.:3.000  3rd Qu.:2.000  3rd Qu.:32.00  3rd Qu.:2.000
##  Max.   :3.000  Max.   :2.000  Max.   :56.00  Max.   :2.000
##      salary
##  Min.   : 57800
##  1st Qu.: 91000
##  Median :107300
##  Mean   :113706
##  3rd Qu.:134185
##  Max.   :231545
```

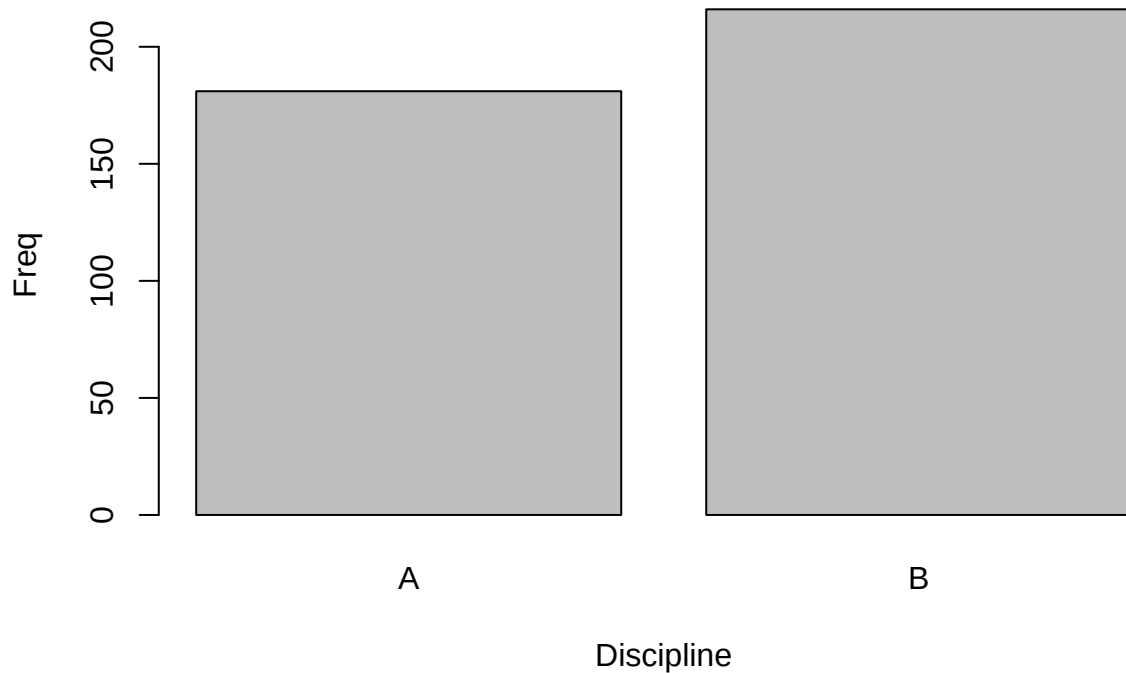
```
# graphing summary
#par(mfrow=c(2, 3))
barplot(table(data$rank), main = "Barplot of Rank", xlab = "Rank", ylab = "Freq",
       names.arg = c("AssistantP", "AssociateP", "Professor"),
       col = c("yellow", "orange", "red"))
```

Barplot of Rank



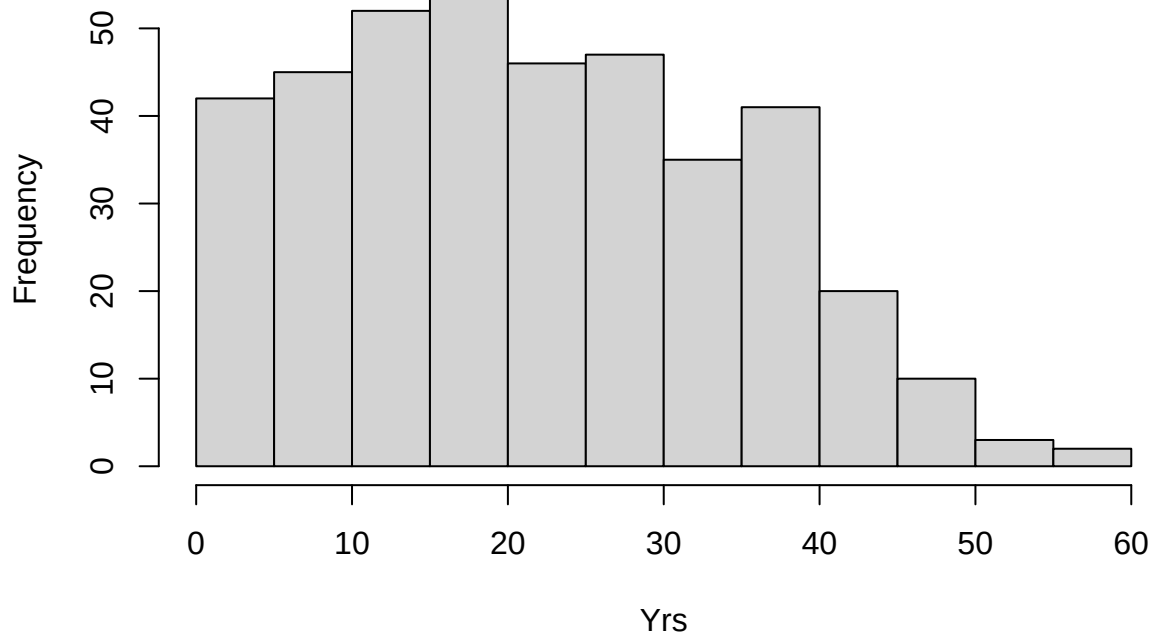
```
barplot(table(data$discipline), main = "Barplot of Disciplines", xlab = "Discipline",  
        ylab = "Freq", names.arg = c("A", "B"))
```

Barplot of Disciplines



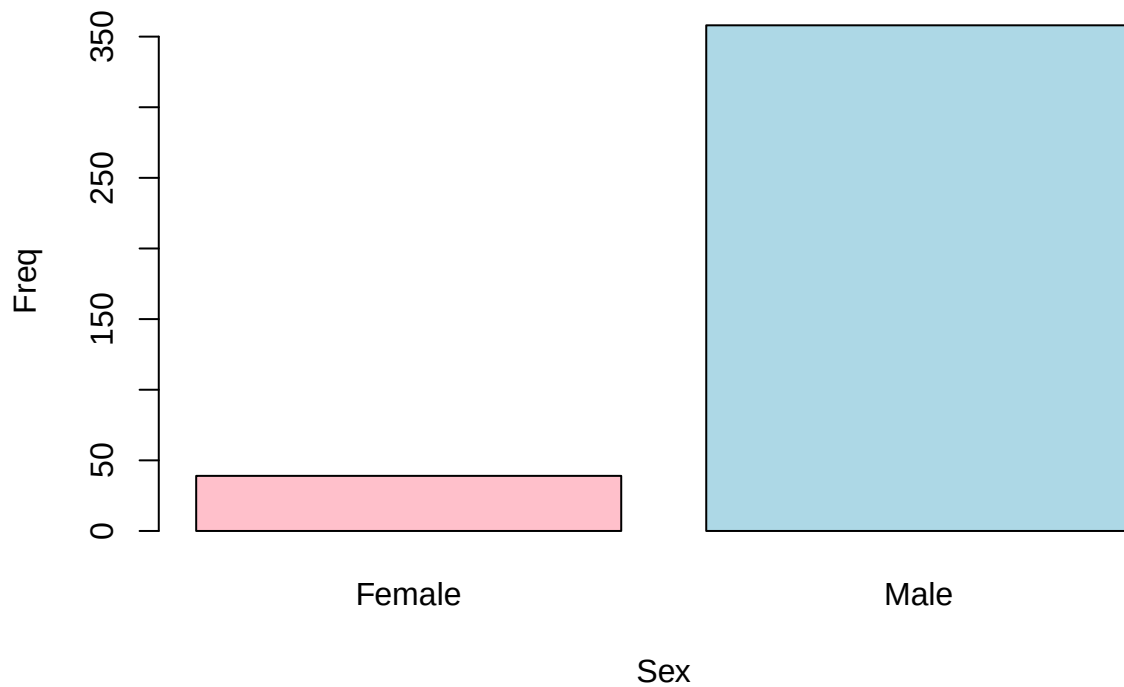
```
hist(data$yrs.since.phd, main = "Histogram of Years Since PHD", xlab = "Yrs")
```

Histogram of Years Since PHD



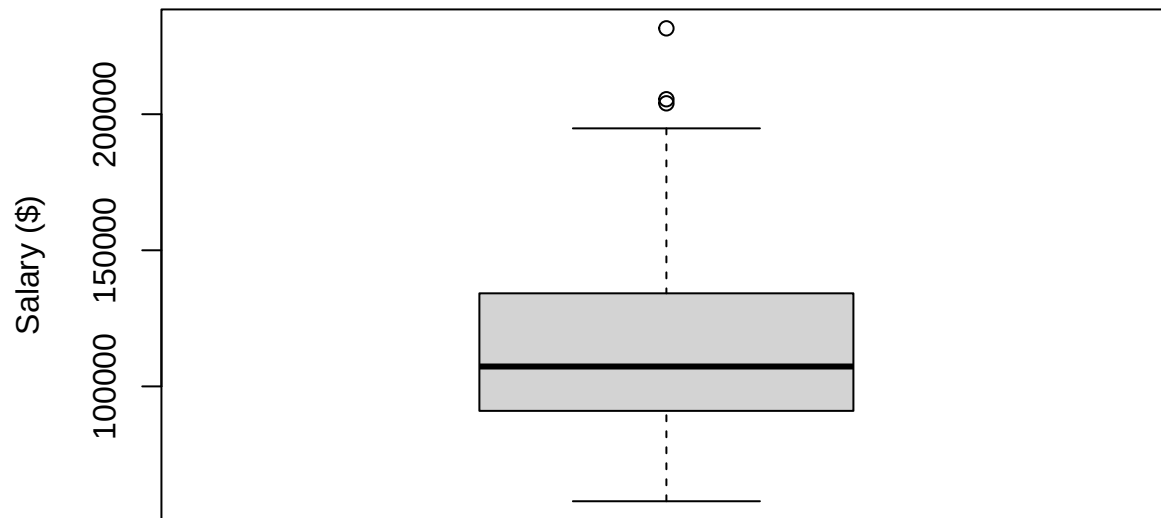
```
barplot(table(data$sex), main = "Stacked Barplot of Sex", xlab = "Sex",  
        ylab = "Freq", names.arg = c("Female", "Male"), col = c("pink", "lightblue"))
```

Stacked Barplot of Sex



```
boxplot(data$salary, main = "Boxplot of Salary", ylab = "Salary ($)")
```

Boxplot of Salary



Answer:

Rank

There are more full professors than assistant or associate professors.

Discipline

Both A and B seem mostly balanced, with B having slightly greater occurrences.

yrs.since.phd

Most staff have between 10 and 40 years of elapsed time since their PhD. Distribution has slight right skew, with very few staff having 50 or more years since their PhD.

Sex

Highly imbalanced, with Males greatly dominating the staff roster.

Salary

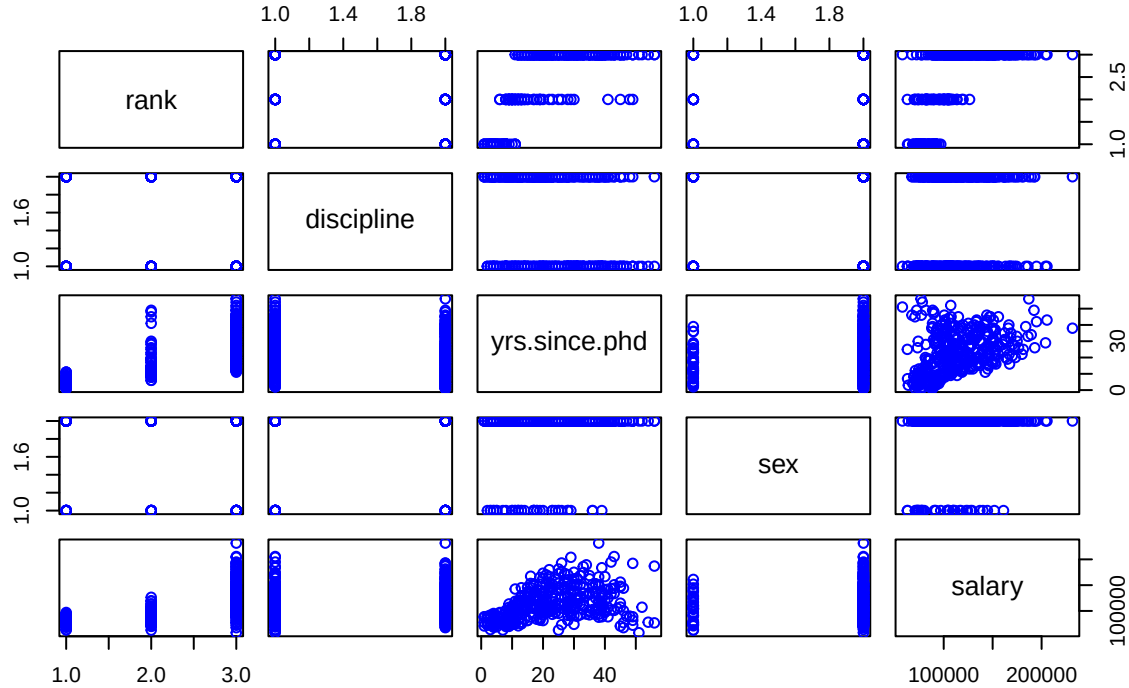
Salaries range from \$57k to \$231k, with a median of \$107k and an average salary of \$113k.

3. Create a scatterplot matrix and briefly describe your findings

Code:

```
pairs(data, main = "Scatterplot", col = "blue")
```

Scatterplot



Answer:

- rank, discipline, sex
These categorical vars are not well represented in a scatterplot.
- yrs.since.phd vs salary
Clear positive correlation, higher salary with more years since phd.
- rank vs salary
Slightly linear association indicating higher rank correlates with higher salary.
- sex vs salary
ANOVA must be used to determine if there is significant difference in salaries between genders.

Model Fitting

- Fit a multiple linear regression model (MLR) with `yrs.since.phd`, `discipline`, `rank`, and `sex` as predictors. Write down the fitted regression equations for each category (e.g., Female, Assistant Professor, theoretical departments). There are 12 categories in total

Code:

```
model <- lm(salary ~ yrs.since.phd + discipline + rank + sex, data = Salaries)
summary(model)
```

```
##
## Call:
## lm(formula = salary ~ yrs.since.phd + discipline + rank + sex,
##     data = Salaries)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -67451 -13860  -1549   10716   97023
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  67884.32    4536.89   14.963 < 2e-16 ***
## yrs.since.phd    61.01     127.01    0.480  0.63124
## disciplineB   13937.47    2346.53    5.940 6.32e-09 ***
## rankAssocProf 13104.15    4167.31    3.145  0.00179 **
## rankProf      46032.55    4240.12   10.856 < 2e-16 ***
## sexMale       4349.37     3875.39    1.122  0.26242
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22660 on 391 degrees of freedom
## Multiple R-squared:  0.4472, Adjusted R-squared:  0.4401
## F-statistic: 63.27 on 5 and 391 DF,  p-value: < 2.2e-16
```

Answer:

- Fitted equation for salary

$$\widehat{\text{salary}} = 67884.32 + 61.01 * (\text{yrs.since.phd}) + 13937.47 * (\text{disciplineB}) + 13104.15 * (\text{AssocProf}) + 46032.55 * (\text{Prof}) + 4349.37 * (\text{Male})$$

1. Female, Assistant, Theoretical

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd})$$

2. Female, Assistant, Applied

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13937.47 \times (\text{disciplineB})$$

3. Female, Associate, Theoretical

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13104.15 \times (\text{AssocProf})$$

4. Female, Associate, Applied

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13104.15 \times (\text{AssocProf}) + 13937.47 \times (\text{disciplineB})$$

5. Female, Full, Theoretical

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 46032.55 \times (\text{Prof})$$

6. Female, Full, Applied

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 46032.55 \times (\text{Prof}) + 13937.47 \times (\text{disciplineB})$$

7. Male, Assistant, Theoretical

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 4349.37 \times (\text{Male})$$

8. Male, Assistant, Applied

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13937.47 \times (\text{disciplineB}) + 4349.37 \times (\text{Male})$$

9. Male, Associate, Theoretical

$$\widehat{\text{salary}} = 67884.32 + 61.01 \times (\text{yrs.since.phd}) + 13104.15 \times (\text{AssocProf}) + 4349.37 \times (\text{Male})$$

10. Male, Associate, Applied

$$\widehat{salary} = 67884.32 + 61.01 \times (yrs.since.phd) + 13104.15 \times (AssocProf) + 13937.47 \times (disciplineB) + 4349.37 \times (Male)$$

11. Male, Full, Theoretical

$$\widehat{salary} = 67884.32 + 61.01 \times (yrs.since.phd) + 46032.55 \times (Prof) + 4349.37 \times (Male)$$

12. Male, Full, Applied

$$\widehat{salary} = 67884.32 + 61.01 \times (yrs.since.phd) + 46032.55 \times (Prof) + 13937.47 \times (disciplineB) + 4349.37 \times (Male)$$

5. State the model assumptions in the previous regression model.

Answer:

- Linear
Dependent var (salary) and independent vars (remaining 4 vars) should have linear relationship.
- Independence
Residuals assumed to be independent, observations should not affect each other.
- Homoscedasticity
Variance of residuals must be constant across independent vars
- Normality of Residuals
Residuals should have a normal distributions.
- No Multicollinearity
Independent vars should not have high correlation with each other.

6. Now fit another MLR with `yrs.since.phd`, `discipline`, `sex` and their interactions. Write down the fitted regression equations for each category

Code:

```
interact_model <- lm(salary ~ yrs.since.phd * discipline * sex, data = Salaries)
summary(interact_model)
```

```
##
## Call:
## lm(formula = salary ~ yrs.since.phd * discipline * sex, data = Salaries)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -83181 -18486  -2710   15015   91016
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    66195.4    11211.2   5.904 7.72e-09 ***
## yrs.since.phd     1306.8      534.3   2.446  0.0149 *
## disciplineB      2677.9    17327.0   0.155  0.8773
## sexMale         22688.4    12136.3   1.869  0.0623 .
## yrs.since.phd:disciplineB    1397.1      928.9   1.504  0.1334
## yrs.since.phd:sexMale     -475.8      557.4  -0.854  0.3939
## disciplineB:sexMale     2194.0    18294.3   0.120  0.9046
## yrs.since.phd:disciplineB:sexMale  -989.3      954.5  -1.036  0.3006
```



```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 26240 on 389 degrees of freedom
## Multiple R-squared:  0.2626, Adjusted R-squared:  0.2493
## F-statistic: 19.79 on 7 and 389 DF,  p-value: < 2.2e-16
```

Answer:

1. Female, Theoretical

$$\widehat{\text{salary}} = 66195.4 + 1306.8 \times (\text{yrs.since.phd})$$

2. Female, Applied

$$\widehat{\text{salary}} = 66195.4 + 1306.8 \times (\text{yrs.since.phd}) + 2677.9 + 1397.1 \times (\text{yrs.since.phd})$$

3. Male, Theoretical

$$\widehat{\text{salary}} = 66195.4 + 1306.8 \times (\text{yrs.since.phd}) + 22688.4 - 475.8 \times (\text{yrs.since.phd})$$

4. Male, Applied

$$\widehat{\text{salary}} = 66195.4 + 1306.8 \times (\text{yrs.since.phd}) + 2677.9 + 22688.4 + 1397.1 \times (\text{yrs.since.phd}) - 475.8(\text{yrs.since.phd}) + 2194$$

Non-linear Regression: An Simulated Example

Suppose the response y depends on the predictor t in the following form:

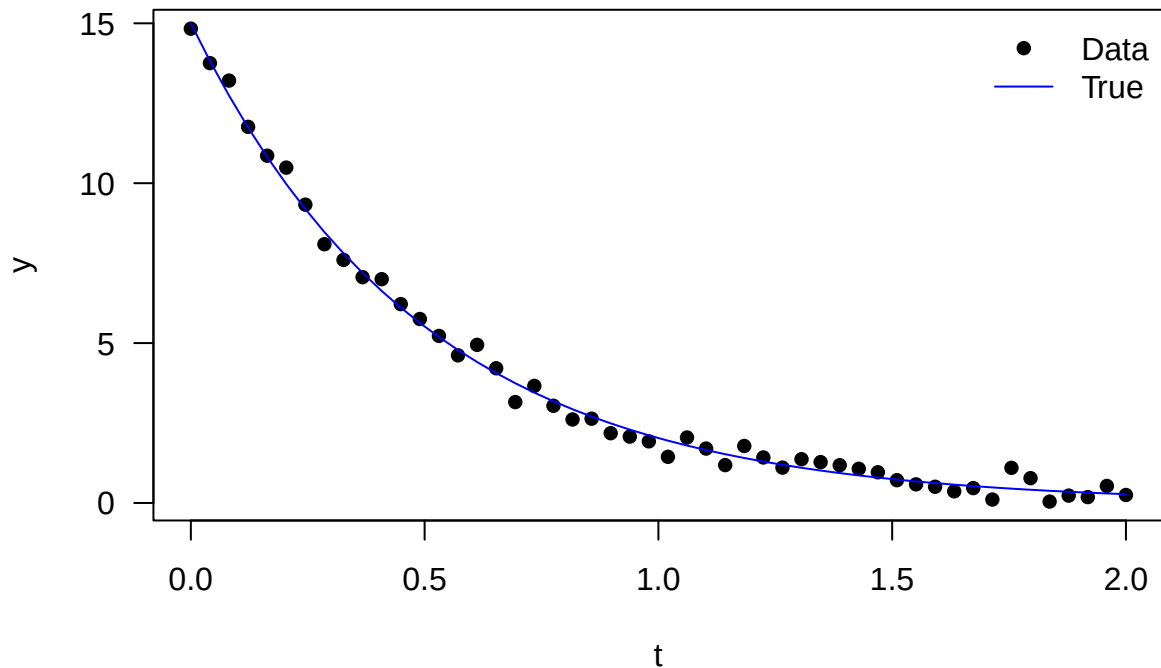
$$y = \alpha \exp(-\beta t) + \epsilon,$$

where $\epsilon \sim N(0, \sigma^2)$, and the true α , β , and σ^2 are 15, 2 and 0.16, respectively. First, let's simulate some data points from this nonlinear model:

Code:

```
alpha = 15; beta = 2; sigma.sq = 0.09
n <- 50
t <- seq(0, 2, len = 50)
set.seed(123)
y <- alpha * exp(-beta * t) + rnorm(n, sd = sqrt(sigma.sq))
data <- data.frame(y = y, t = t)

plot(t, y, las = 1, pch = 16)
lines(t, alpha * exp(-beta * t), type = "l", col = "blue")
legend("topright", legend = c("Data", "True"), pch = c(16, NA), lty = c(NA, 1),
      col = c("black", "blue"), bty = "n")
```



7. Use the `nls` function to obtain nonlinear least-squares estimates $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\sigma}^2$. To use `nls`, provide formula = `y ~ alpha * exp(-beta * t)`, start = `list(alpha = alpha_0, beta = beta_0)`, where `alpha_0` and `beta_0` are initial guesses of the parameters α and β

Code:

```
alpha_0 <- 10
beta_0 <- 1

nonlinear_fit <- nls(y ~ alpha * exp(-beta * t), data = data,
                    start = list(alpha = alpha_0, beta = beta_0))
summary(nonlinear_fit)

##
## Formula: y ~ alpha * exp(-beta * t)
##
## Parameters:
##      Estimate Std. Error t value Pr(>|t|)
## alpha  15.0962    0.1484  101.72  <2e-16 ***
## beta    2.0113    0.0293   68.65  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2796 on 48 degrees of freedom
##
## Number of iterations to convergence: 4
## Achieved convergence tolerance: 9.237e-09

alpha_hat <- coef(nonlinear_fit)['alpha']
beta_hat <- coef(nonlinear_fit)['beta']

# residual variace
res <- residuals(nonlinear_fit)
sigma_hat_sq <- var(res)
```

```

# estimated vals
print(c("Estimated alpha:", alpha_hat))

##                                alpha
## "Estimated alpha:" "15.0962242708225"

print(c("Estimated beta:", beta_hat))

##                                beta
## "Estimated beta:" "2.01131765084923"

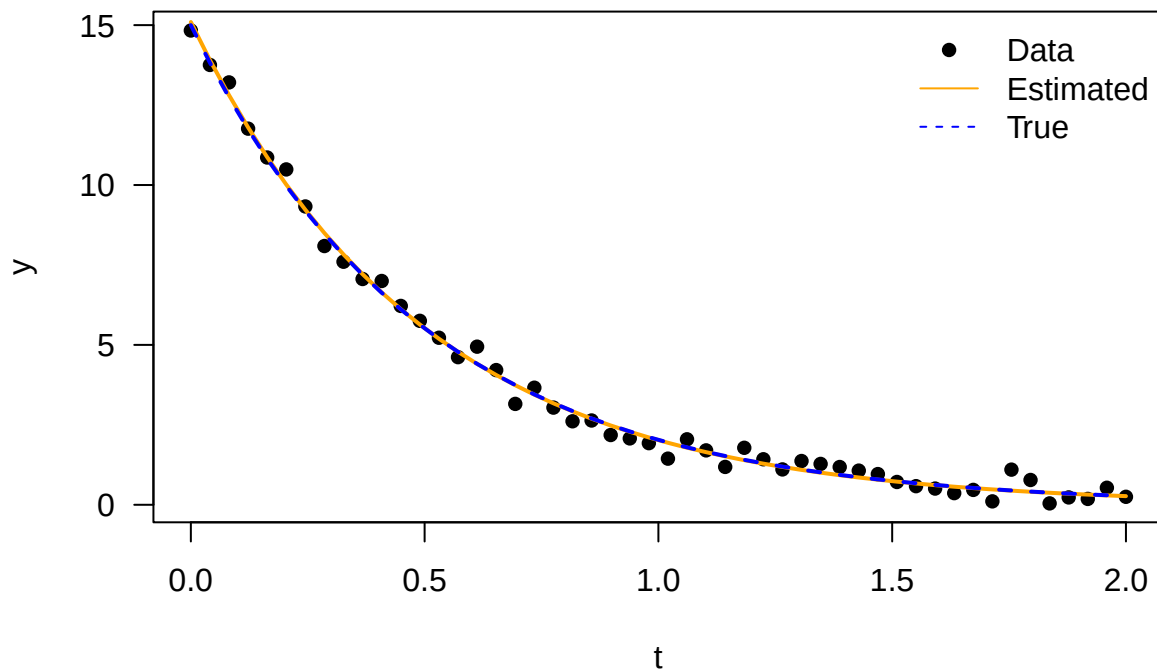
print(c("Estimated sigma squared:", sigma_hat_sq))

## [1] "Estimated sigma squared:" "0.0765514384068425"

# plotting estimated curve
plot(t, y, las = 1, pch = 16, main = "Nonlinear Least Squares Fit")
lines(t, alpha_hat * exp(-beta_hat * t), col = "orange", lwd = 2)
lines(t, 15 * exp(-2 * t), col = "blue", lwd = 2, lty = 2) # to compare
legend("topright", legend = c("Data", "Estimated", "True"),
      pch = c(16, NA, NA), lty = c(NA, 1, 2),
      col = c("black", "orange", "blue"), bty = "n")

```

Nonlinear Least Squares Fit



Answer:

Estimated values:

$$\hat{\alpha} = 15.0962242708225, \hat{\beta} = 2.0113176, \hat{\sigma}^2 = 0.0765514384$$

Estimate equation:

$$\hat{y} = 15.0962e^{-2.011376t} + \epsilon, \quad \epsilon \sim N(0, 0.07655)$$

8. Write down the fitted equation and the estimated variance $\hat{\sigma}^2$

Answer:

Fitted Equation $\hat{y} = 15.0962e^{-2.011376t} + \epsilon, \quad \epsilon \sim N(0, 0.07655)$

Estimated Variance $\hat{\sigma}^2 = 0.07655$

9. Apply the natural log transformation to the simulated response, then fit a simple linear regression.
Back-transform to obtain the fit on the original scale

Code:

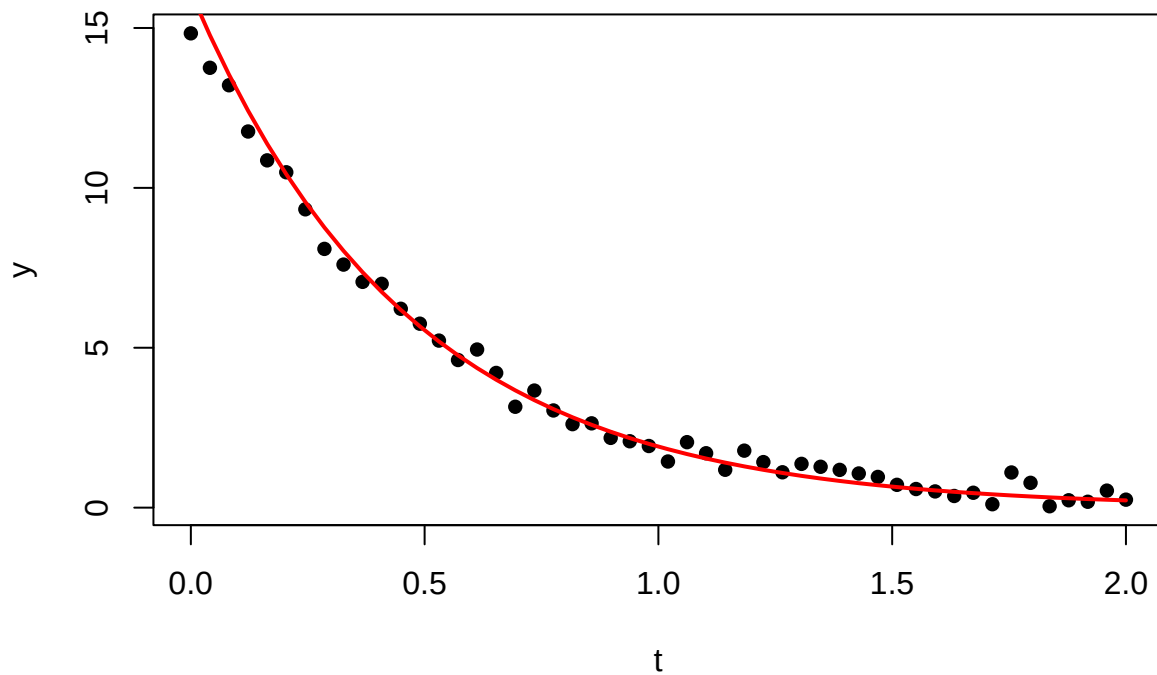
```
# simple linear reg
linear_fit <- lm(log(y) ~ t, data = data)
summary(linear_fit)

##
## Call:
## lm(formula = log(y) ~ t, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.98777 -0.06917 -0.00311  0.10335  1.05859
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.7798      0.1233   22.55  <2e-16 ***
## t             -2.1332      0.1062  -20.09  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4423 on 48 degrees of freedom
## Multiple R-squared:  0.8937, Adjusted R-squared:  0.8915
## F-statistic: 403.5 on 1 and 48 DF,  p-value: < 2.2e-16

# coeff
alpha_log <- exp(coef(linear_fit)[1])
beta_log <- -coef(linear_fit)[2]

# back transform
plot(t, y, pch = 16, main = "Log-Transformed Linear Fit (Back-Transformed)")
lines(t, alpha_log * exp(-beta_log * t), col = "red", lwd = 2)
```

Log-Transformed Linear Fit (Back-Transformed)



Answer:

- Nonlinear Model

$$\ln(y) = \ln(\alpha) - \beta t + \epsilon', \epsilon' \sim N(0, \sigma'^2)$$

10. Comparing the nonlinear regression method and the linear regression with log-transformed response, which method would you prefer in this example? Explain your answer

Answer:

- NLS as it can give parameter estimates with greater accuracy without being affected by transformations.
-